

JEE MAIN REVISION MASTER: FLUID MECHANICS

2026 2 QUESTIONS 3.33% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 12 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Fluid Mechanics - Core Application	12 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Fluid Mechanics - Core Application (High Yield — 12 PYQs)

Core Principle: Key problem pattern in Kcet Fluid Mechanics.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A 30 cm long capillary tube is dipped in water, water rises upto a height of 10 cm due to capillarity. If this experiment is conducted in ..." [Ans: C]
- [2026 Shift] Q#2 (MCQ): "Instrument fitted in the carburetor of the automobile to provide the correct mixture of air and fuel necessary for combustion works on..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "A horizontal pipe carries water in a streamlined flow. At a point along the pipe, where the cross-sectional area is 10cm^2 , the veloci..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A 30 cm long capillary tube is dipped in water, water rises upto a height of 10 cm due to capillarity. If this experiment is conducted in a freely falling water in an elevator, then the length of the water column becomes

- (A) 10 cm
- (B) 20 cm
- (C) 30 cm **[Correct]**
- (D) Zero

Correct Answer: C

Solution Analysis: In normal conditions, water rises in a capillary tube because the upward force due to surface tension balances the weight of the water column. The capillary rise is $h = \frac{2T \cos \theta}{\rho g r}$ Here, g is the acceleration due to gravity. In a freely falling elevator, the effective acceleration due to gravity becomes zero: $g_{\text{eff}} = 0$ So the water column has no effective weight. Since there is no downward weight to balance the upward pull due to surface tension, the water will rise as much as possible in the tube. But the tube is only 30 cm long. So the water can fill the whole tube.